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Digital Forensics

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Table of Contents

1	Digital Forensics Summary	2
1.1	Principles of digital forensics	2
1.2	Digital multimedia forensics: definition, classification.	2
1.2.1	Idea	2
1.3	Active and passive approaches for multimedia authentication.	2
1.3.1	Active approach:	2
1.3.2	Passive and blind approach: multimedia forensics	3
1.4	Image generation pipeline	3
1.5	Camera identification based on photo response non-uniformity (PRNU) .	3
1.6	Media forensics based on in-camera fingerprints.	4
1.6.1	Source identification	4
1.6.2	Usage of digital fingerprints	4
1.7	Application of geometrical fingerprints in digital forensics.	4
1.8	Out-camera fingerprints in digital forensics.	4
1.9	Principles of image JPEG compression.	4
1.10	JPEG compression: general compression scheme	5
1.10.1	Lossless compression	5
1.11	JPEG compression: sources of redundancies	5
1.12	JPEG compression: particularities of luminance and chrominance channels processing	5
1.13	JPEG compression artefacts	5
1.14	Detection of double JPEG compression	5
1.14.1	Double quantization: footprints	6
1.15	Image editing forensics: inconsistency in light	6
1.16	Image editing forensics: local filtering traces	6
1.17	Photo manipulation with inconsistent shadows: basic assumptions	6
1.18	Photo manipulation with inconsistent shadows: cast shadow	7
1.19	Photo manipulation with inconsistent shadows: attached shadow	7
1.20	Photo manipulation with inconsistent shadows: mathematical model . . .	7
1.21	Traces of image editing: resampling detection	7
1.22	Traces of image editing: detection of global contrast enhancement	7
1.23	Traces of image editing: detection of local contrast enhancement	7

1 Digital Forensics Summary

1.1 Principles of digital forensics

Forensic Gist: The systematic process for investigating digital evidence while maintaining legal admissibility.

- **Acquisition:** Securely preserve digital evidence without alteration using forensic tools
- **Identification:** Locate and extract relevant data, including hidden or deleted content
- **Evaluation:** Analyze extracted data to reconstruct events and establish timelines
- **Presentation:** Communicate findings clearly to both technical and non-technical audiences

Key Users: Law enforcement, legal teams, corporations investigating internal matters

1.2 Digital multimedia forensics: definition, classification.

Forensic Gist: Verifying the authenticity of digital media (images, video, audio) by detecting manipulation traces.

Core Problem: Digital media can no longer be automatically trusted as evidence due to: - Easy manipulation tools - Social impact of misinformation - Scientific challenge of establishing ultimate reliability

1.2.1 Idea

Forensic Approach: Analyze inherent “digital fingerprints” left during: - Creation (camera-specific patterns) - Processing (editing software traces) - Scene characteristics (lighting, geometry)

Digital fingerprints Three main fingerprint categories: - **In-camera fingerprints:** Hardware/software signatures unique to specific devices - **Out-camera fingerprints:** Statistical changes from editing operations - **Scene fingerprints:** Physical consistency checks (lighting, perspective, shadows)

1.3 Active and passive approaches for multimedia authentication.

1.3.1 Active approach:

Forensic Gist: Proactive protection using embedded security features.

Methods: - **Cryptographic Signature:** Digital signatures created at source and verified later - **Digital Watermarking:** Invisible data embedded for authentication

Disadvantages Limitations of Active Methods: - Heavy computational requirements - Cannot distinguish compression from malicious tampering - Protection disappears after decryption - Requires pre-existing infrastructure

1.3.2 Passive and blind approach: multimedia forensics

Forensic Gist: Analyze media without pre-embedded markers to answer: - **Source identification:** Which device created this? - **Integrity verification:** Has this been altered?

Advantages Benefits of Passive Methods: - No special preparation required - Works with any existing digital media - No need for device databases or standards

1.4 Image generation pipeline

Forensic Gist: Understanding camera processing steps reveals detectable fingerprints.

Processing Chain: 1. **Light projection** onto sensor grid 2. **Amplification and quantization** of electrical signals 3. **Signal adjustment** for white balance 4. **Color filtering** through Bayer pattern CFA 5. **Interpolation/demosaicking** to reconstruct full color 6. **Color/gamma correction** for display 7. **Sharpening/denoising** filters 8. **Storage** in compressed format

Each step leaves forensic traces that can identify the device or detect edits.

1.5 Camera identification based on photo response non-uniformity (PRNU)

Forensic Gist: Use sensor manufacturing imperfections as unique camera fingerprints.

Sensor Noise Components: - **Fixed Pattern Noise (FPN):** Additive noise visible in dark conditions - **Photo-Response Non-Uniformity (PRNU):** Multiplicative noise from pixel sensitivity variations

Mathematical Model: $I(x, y) = I_0(x, y) + I_0(x, y) \cdot K(x, y) + N(x, y)$

Where: - $I(x, y)$: Observed pixel value - $I_0(x, y)$: Ideal noise-free image - $K(x, y)$: PRNU pattern (camera fingerprint) - $N(x, y)$: Random noise

Forensic Application: Estimate $K(x, y)$ by averaging multiple images from same camera, then compare unknown images against this fingerprint.

1.6 Media forensics based on in-camera fingerprints.

1.6.1 Source identification

Forensic Gist: Determine capture device characteristics through inherent signatures.

Identification Hierarchy: - Device class (CCD vs CMOS) - Brand/model specifications - Individual device identification

1.6.2 Usage of digital fingerprints

Applications: - **Source identification:** Match against reference device databases -

Forgery detection: Identify fingerprint inconsistencies indicating edits

Tampering detection Specific Techniques: - Analyze Color Filter Array (CFA) interpolation patterns - Detect processing chain anomalies (white balance, sharpening traces) - Compare against expected camera model behaviors

1.7 Application of geometrical fingerprints in digital forensics.

Forensic Gist: Use physical scene properties to detect inconsistencies.

Key Methods: - **Copy-move detection:** Find duplicated regions using sliding window analysis - **Highlight analysis:** Model eyes as spheres to verify light source consistency - **Shadow analysis:** Check lighting direction coherence across objects - **Chromatic aberration:** Detect cut-paste operations through lens distortion patterns

1.8 Out-camera fingerprints in digital forensics.

Forensic Gist: Detect post-capture processing through statistical alterations.

Detectable Operations: - Compression artifacts - Geometric transformations (rotation, scaling) - Filter applications (blur, sharpening) - Contrast adjustments - Multiple processing combinations

1.9 Principles of image JPEG compression.

Forensic Gist: Understand compression artifacts that serve as forensic markers.

JPEG Advantages: - Effective for photographic content with smooth variations - Exploits human visual system limitations - Separates luminance (detail) from chrominance (color) processing

Redundancy Types: - **Statistical redundancy:** Correlations between neighboring pixels - **Psychovisual redundancy:** Information humans don't perceive well

1.10 JPEG compression: general compression scheme

Forensic Gist: Compression pipeline creates detectable patterns.

Processing Steps: 1. **Transform (T):** Convert to frequency domain (DCT) 2. **Quantize (Q):** Reduce precision (lossy step) 3. **Code (C):** Compress losslessly (Huffman coding)

Key Operation: $F_{quantized}[u, v] = \text{round}\left(\frac{F[u, v]}{Q(u, v)}\right)$

Where $Q(u, v)$ is quantization matrix values controlling quality/compression trade-off.

1.10.1 Lossless compression

Alternative: Remove quantizer step for perfect reconstruction but limited compression.

1.11 JPEG compression: sources of redundancies

Forensic Gist: Different redundancy types leave distinct compression footprints.

Spatial: Pixel neighborhood correlations

Spectral: Color channel relationships

Temporal: Frame-to-frame video similarities

Human Perception Limits:

High frequency detail, color subtlety, rapid changes - all can be compressed without noticeable quality loss.

1.12 JPEG compression: particularities of luminance and chrominance channels processing

Forensic Gist: Separate processing of brightness vs. color creates forensic signatures.

YCbCr Color Space: - Y (luminance): Preserved at full resolution - Cb, Cr (chrominance): Subsampled (typically 4:2:0)

Data Reduction Example: Original: $3 \times 64^2 = 12,288$ values
JPEG: $64^2 + 2 \times 32^2 = 6,144$ values (50% reduction)

1.13 JPEG compression artefacts

Forensic Gist: Compression imperfections serve as authenticity indicators.

Visible Artifacts: - **Blocking:** 8×8 block boundary discontinuities - **Ringings:** Ghosting near edges from frequency truncation - **Graininess:** Quantization noise appearing as dots - **Blurring:** High frequency detail loss

1.14 Detection of double JPEG compression

Forensic Gist: Multiple compressions indicate possible image manipulation.

Forensic Importance: Double JPEG compression often occurs when edited images are resaved.

1.14.1 Double quantization: footprints

Mathematical Analysis: First quantization: $Q_1(x) = \text{round}\left(\frac{x}{a}\right)$

Second quantization: $Q_2(x) = \text{round}\left(\frac{Q_1(x) \cdot a}{b}\right)$

Statistical Patterns: - When $a < b$: Histograms show periodic empty bins (“holes”)
- When $a > b$: Histograms show periodic bin overcrowding

Detection Method: Analyze DCT coefficient histograms for periodic patterns detectable via Fourier analysis.

1.15 Image editing forensics: inconsistency in light

Forensic Gist: Detect copy-paste forgeries through lighting direction mismatches.

Key Assumptions: - Lambertian surface reflectance - Constant object material properties - Distant light source approximation

Forensic Application: Objects from different images will likely have incompatible lighting conditions.

1.16 Image editing forensics: local filtering traces

Forensic Gist: Detect editing operations through filter characteristic patterns.

Median Filtering Detection: Identifies unnatural structural uniformity **Unsharp Masking Detection:** Reveals overshoot artifacts at edges

Both filters are commonly used to hide manipulation traces or enhance manipulated regions.

1.17 Photo manipulation with inconsistent shadows: basic assumptions

Forensic Gist: Use shadow physics to detect composite images.

Shadow Types: - **Cast shadows:** Objects blocking light onto surfaces - **Attached shadows:** Self-shading on object surfaces

Core Principle: Single light source should produce consistent shadow directions across entire scene.

1.18 Photo manipulation with inconsistent shadows: cast shadow

1.19 Photo manipulation with inconsistent shadows: attached shadow

Forensic Gist: Different shadow types provide complementary authenticity checks.

Cast Shadows: Define light source position through shadow-object relationships

Attached Shadows: Determine lighting direction from surface gradient analysis

1.20 Photo manipulation with inconsistent shadows: mathematical model

Forensic Gist: Formalize shadow consistency checking as geometric constraint problem.

Mathematical Framework: For each shadow constraint: $n \cdot x - n \cdot p \geq 0$

System Solution: Solve $Nx - P \geq -s$ where: - N : Constraint normals matrix - P : Reference points vector - s : Slack variables (nonzero values indicate inconsistencies)

Decision Rule: All $s = 0 \rightarrow$ consistent shadows; Any $s > 0 \rightarrow$ potential forgery

1.21 Traces of image editing: resampling detection

Forensic Gist: Detect geometric transformations (resizing, rotation) common in forgeries.

Technical Basis: Resampling creates periodic pixel correlations through interpolation: $y = Hx$

Where H is interpolation matrix creating linearly dependent relationships detectable via Fourier analysis of probability maps.

1.22 Traces of image editing: detection of global contrast enhancement

Forensic Gist: Identify overall image adjustments through histogram and frequency analysis.

Detection Methods: - Histogram analysis for isolated peaks/gaps - Frequency domain energy distribution checking - Expected low-pass signal violation detection

1.23 Traces of image editing: detection of local contrast enhancement

Forensic Gist: Locate specific tampered regions through block-based analysis.

Methodology: Divide image into blocks (200×200 to 20×20 pixels) and apply global techniques locally to identify manipulated areas.